ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Mechatronics, levers and belt drives; series circuits and logic with switches; pneumatic circuits, symbols and cylinder force; start/stop control with indicator lamps; workshop safety, PPE and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw all symbols and circuits neatly, using a ruler.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

QUESTION 1

1.1 Define the term *mechatronics*. (2)

1.2 FIGURE 1 below shows a lever used to lift a load.

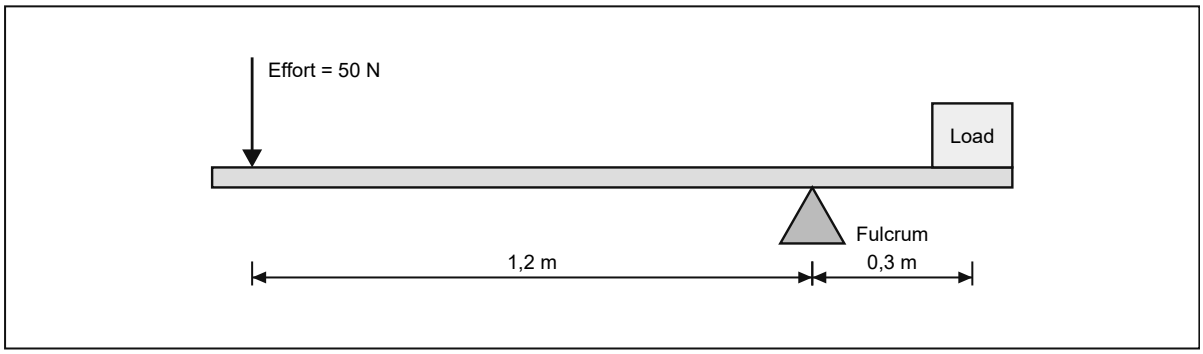


FIGURE 1

1.2.1 Identify the class of the lever. (1)

1.2.2 Calculate the load that the lever can lift. (3)

1.2.3 Calculate the mechanical advantage of the lever. (2)

1.3 A driver pulley with a diameter of 100 mm turns at 1 440 r/min. It drives a pulley with a diameter of 300 mm by means of a V-belt.

1.3.1 Calculate the speed of the driven pulley. (3)

1.3.2 Calculate the velocity ratio of the belt drive. (2)

1.4 State TWO advantages of a gear drive over a belt drive. (2 × 2) (4)

1.5 State the energy change that takes place in EACH of the following:

1.5.1 Solar panel (1)

1.5.2 Loudspeaker (1)

1.5.3 Electric heater (1)

[20]

QUESTION 2

2.1 State Ohm's law. (2)

2.2 Three resistors of 10 Ω, 20 Ω and 30 Ω are connected in series across a 12 V supply. Calculate the following:

2.2.1 The total resistance (2)

2.2.2 The current in the circuit (2)

2.2.3 The voltage across the 30 Ω resistor (2)

2.3 Draw a circuit in which two switches, A and B, and a lamp E1 form an OR function. Also draw the truth table of the circuit. (6)

2.4 Draw the symbol of EACH of the following:

2.4.1 Fuse (2)

2.4.2 Normally closed (NC) contact (2)

2.5 Write the following abbreviations in full:

2.5.1 LED (1)

2.5.2 AC (1)

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QUESTION 3

3.1 Explain the function of the air receiver in a compressed-air system. (3)

3.2 FIGURE 2 below shows a pneumatic circuit.

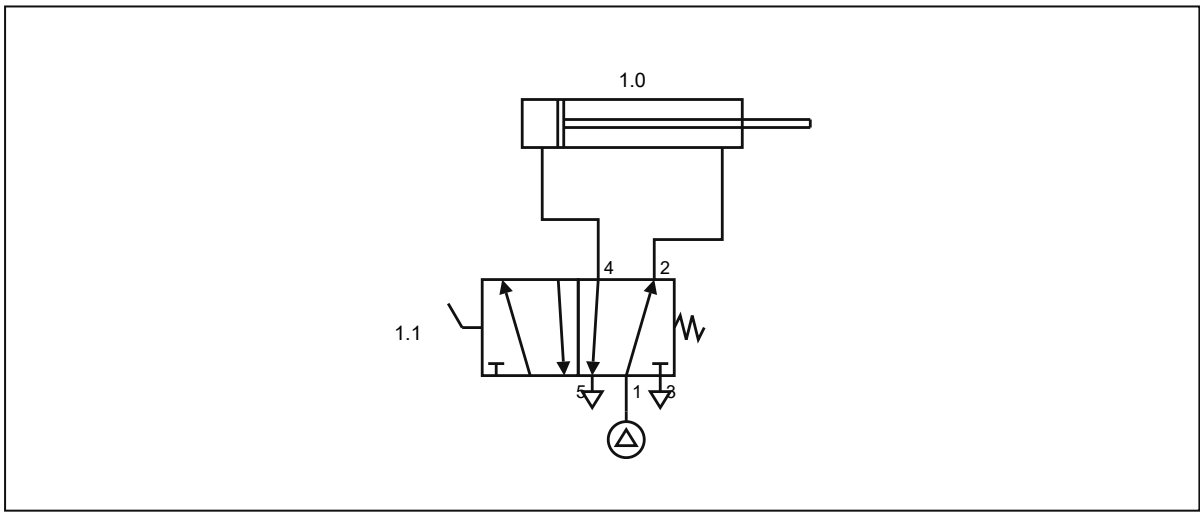


FIGURE 2

3.2.1 Name the components labelled 1.0 and 1.1. (2)

3.2.2 Explain what happens when the lever of valve 1.1 is operated. (3)

3.2.3 Explain what happens when the lever of valve 1.1 is released. (2)

3.2.4 Name the component that can be added to the circuit to make the cylinder extend more slowly. (2)

3.3 Draw the symbol of a 3/2-way valve (normally closed) that is operated by a push button and has a spring return. (4)

3.4 A double-acting cylinder has a piston diameter of 50 mm and is supplied with air at 500 kPa. Calculate the force when the cylinder extends. (4)

[20]

QUESTION 4

4.1 Explain the function of a contactor in a motor control circuit. (2)

4.2 Explain the difference between an isolator and a circuit breaker. (2 × 2) (4)

4.3 Draw a control circuit in which a motor contactor K1 is started with a START button (S2) and stopped with a STOP button (S1). The circuit must have a holding contact and an overload contact (F1). A green lamp H1 must show that the motor is running and a red lamp H2 must show that the motor is stopped. (10)

4.4 Why must a motor control circuit have overload protection? (2)

4.5 State the colour of an emergency stop button and of the background behind it. (2)

[20]

QUESTION 5

5.1 List FOUR general safety rules that must be followed in a mechatronics workshop. (4)

5.2 Name the personal protective equipment (PPE) that must be worn for EACH of the following tasks:

5.2.1 Grinding a metal part (1)

5.2.2 Handling sharp sheet metal (1)

5.2.3 Working next to a noisy compressor (1)

5.2.4 Soldering (1)

5.3 A conveyor motor does not start when the START button is pressed. List, in a logical order, FIVE checks that you would make. (5)

5.4 Explain the difference between a routine inspection and an overhaul. (2 × 2) (4)

5.5 Give THREE reasons why records of faults and repairs must be kept. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER